

Galois Theory

Why quintic equations have **no solution formula**

Measuring the solvability of equations in the language of symmetry

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Goals of This Deck

- 1 What is a Galois group** Capturing as a group the symmetry that preserves relations (sums, products) among roots
- 2 Group shrinks, field grows** Base Q , intermediate L , splitting K correspond to groups
- 3 Solvability condition** chain reaching $\{e\}$ = solvable by radicals
- 4 The degree-5 wall** The special nature of A_5 rules out a formula

The main part is fairly abstract — reading the **appendix (worked examples)** first makes it easier to follow. The common early stumbling points are collected in **Appendix 3**.

● STEP 1

Galois group = permutations preserving relations (sums, products) among roots

For any rational expression r whose value built from the roots is **rational**, a permutation σ of the roots keeps the value **rational** — collecting all such permutations gives a group: the Galois group G .

$$\forall r, \forall \sigma \in G : r(\alpha_1, \dots, \alpha_n) \in \mathbb{Q} \Rightarrow \sigma(r(\alpha_1, \dots, \alpha_n)) \in \mathbb{Q}$$

Note: imposing this for **every** r yields the same group as requiring the value itself be preserved: $\sigma(r(\alpha_1, \dots, \alpha_n)) = r(\alpha_1, \dots, \alpha_n)$.

Simple examples — rationals stay fixed, roots swap

$$x^3 - 1 = 0$$

Roots: $1, \omega, \omega^2$ (ω : cube root of 1).

- The rational 1 cannot move (fixed)
- Swapping $\omega \leftrightarrow \omega^2$ preserves values → OK

Galois group = cyclic group C_2 (order 2)

$$x^4 - 4x^2 + 1 = 0$$

$\alpha_1 = (\sqrt{6} + \sqrt{2})/2, \alpha_2 = (\sqrt{6} - \sqrt{2})/2, \alpha_3 = -\alpha_1, \alpha_4 = -\alpha_2$

Relations: $\alpha_1 \cdot \alpha_2 = 1, \alpha_1 + \alpha_3 = 0$ (both rational)

Swapping only $\alpha_1 \leftrightarrow \alpha_2$...

- $\alpha_1 \cdot \alpha_2 = 1$ is preserved (just reordering)
- but $\alpha_1 + \alpha_3 = 0$ becomes $-\sqrt{2}$, broken → NG

Swap $\alpha_3 \leftrightarrow \alpha_4$ together with $\alpha_1 \leftrightarrow \alpha_2$ → both preserved, OK

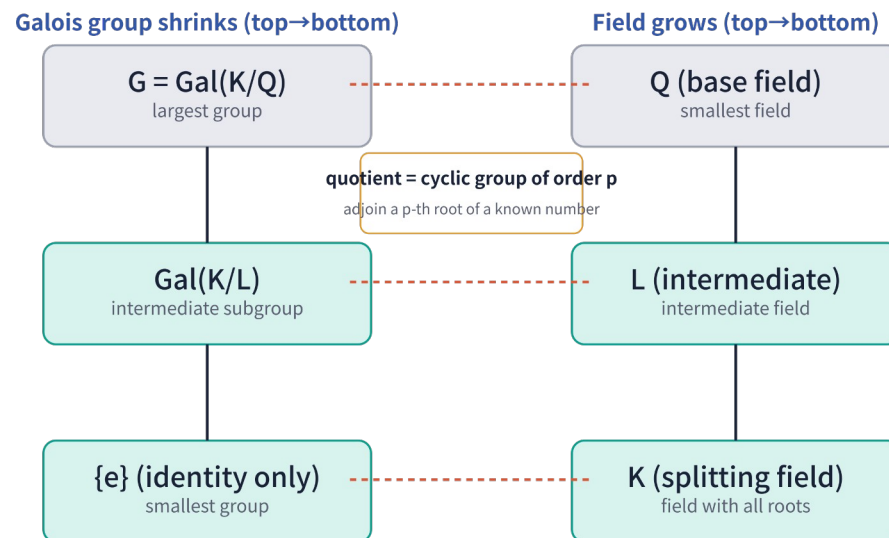
● STEP 2

As the Galois group shrinks, the field grows

Base $Q \subset$ intermediate $L \subset$ splitting field K . Each intermediate L corresponds to the subgroup $\text{Gal}(K/L)$ fixing L :
 $L \mapsto \text{Gal}(K/L) = \text{Aut}(K/L) = \{ \sigma \in G : \sigma(x) = x \ (\forall x \in L) \}$

As the group shrinks the field grows; the elements adjoined to the field are fixed over L inside K .

At each step the quotient is a cyclic group of order p (p prime), and the adjoined element is a p -th root of a previously known number.
Premise: K contains all conjugates of the adjoined elements (Galois extension). At each step, order of the quotient group = degree of that one field extension, so group shrinking and field growth match exactly.



● STEP 3

When an equation is solvable

An equation being solvable by radicals (the four operations and n -th roots) is the same as the Galois group shrinking in the following way:

the quotient at each step is cyclic — a chain from G down to $\{e\}$

shrinking chain

shrink the group step by step

cyclic quotient

each step divides cleanly

reaching $\{e\}$

down to the identity only

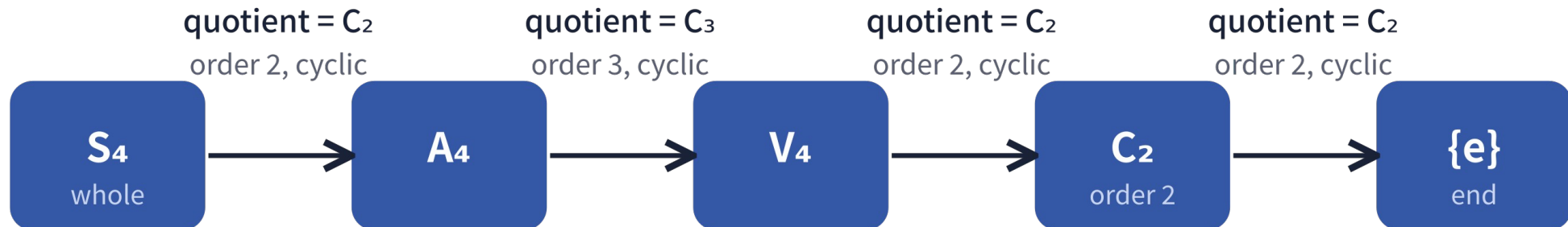
Note

- "Each quotient is cyclic" means the same as "each step is a normal subgroup with abelian quotient (which can be refined so each quotient is cyclic)."
- Definition of a solvable group: one that shrinks to $\{e\}$ via such a chain (with abelian quotients).

- $n \leq 4$

Up to degree 4 — the chain runs all the way

S_4 has a shrinking chain whose quotients are all cyclic. So formulas exist for degree 2, 3, and 4 equations.

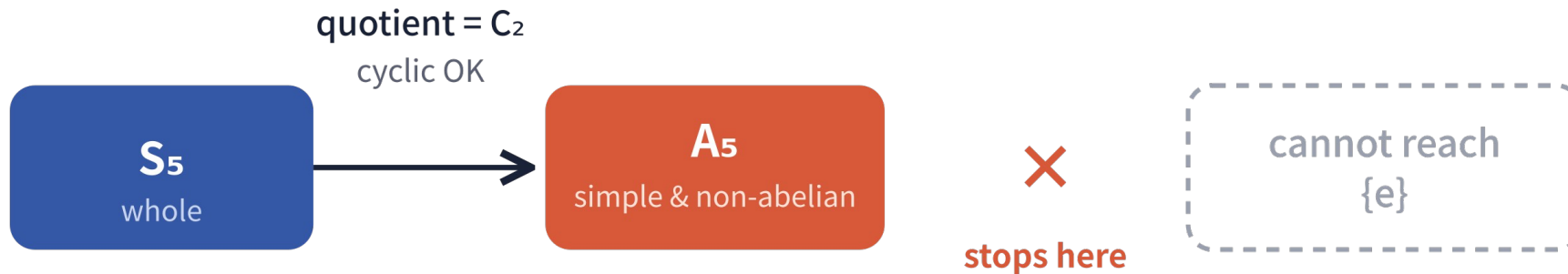


Every quotient is cyclic → reaches {e} → quartics are solvable

● $n \geq 5$

Degree 5 and up — stuck at A_5

A_5 is a “simple group”: the only subgroups usable to shrink A_5 one step (by taking a quotient) are $\{e\}$ and A_5 itself. A_5 does have other subgroups, but dividing by them yields no group, so they cannot be used to shrink it.



No quotient can properly shrink A_5
So no general formula exists for degree ≥ 5

Why it stops at A_5

Image: an indivisible number

A prime is divisible only by 1 and itself.

A simple group

is similar: the smallest unit, divisible only by $\{e\}$ and itself.

A_5 ($n \geq 5$) is

an indivisible block that is not even cyclic. So it cannot in principle be split into cyclic steps.

Two options to shrink A_5

group before / group after = quotient (factor group)

The only usable “after” groups for A_5 are **$\{e\}$ and A_5 itself**.

Group after = $\{e\} \rightarrow$ quotient **$A_5/\{e\} = A_5$**

But A_5 is non-abelian, so it is not cyclic.

Group after = A_5 itself $\rightarrow$ quotient **$A_5/A_5 = \{e\}$**

This quotient is the trivial (cyclic) group, but it just goes $A_5 \rightarrow A_5$, so nothing shrinks.

Conclusion

$$n \leq 4$$

the chain reaches $\{e\} \rightarrow$ a solution formula exists

$$n \geq 5$$

A_n becomes simple and non-abelian, so no step can shrink S_n further.

The chain stops partway and never reaches $\{e\} \rightarrow$ **no solution formula exists**

Abel–Ruffini theorem: no general formula using the four operations and radicals exists for equations of degree 5 or higher.

The skeleton of the argument and the pillars of Galois theory

Argument block	Corresponding pillar	Page
Characterizing the Galois group	permutation = automorphism; the whole set = the Galois group	P3
Group shrinks, field grows	field $\rightarrow$ group correspondence Galois extension (field containing all conjugates of the adjoined elements) order = degree links group and field cyclic extension = adjoining a radical	P4
Solvability (shrinking chain)	solvable group $\Leftrightarrow$ solvable by radicals	P5
Conclusion (breaks at A_5)	simplicity of $A_5 \rightarrow$ unsolvable	P7–9

One-page recap

Capture symmetry as a group

the Galois group = permutations preserving the value of rational expressions built from the roots

Field and group correspond inversely

the larger the field, the smaller the group

Solvable = the chain reaches $\{e\}$

can it shrink to $\{e\}$ with cyclic quotients at each step

The degree-5 wall is A_5

simple and non-abelian, so the chain breaks

Appendix

What it means to solve an equation with radicals and arithmetic

Group shrinking and field growth, through a concrete example

● Appendix 1

Solving equations by defining new roots

In the rationals (field $\mathbb{Q}$), the roots $\sqrt{2}$, $-\sqrt{2}$ of $x^2-2=0$ do not exist. To solve it we must define (adjoin) $\sqrt{2}$.

Adjoining $\sqrt{2}$ lets us write $(x-\sqrt{2})(x+\sqrt{2})=0$, fixing the roots. Over $\mathbb{Q}$ we know there are two roots but cannot tell their values apart: $\sqrt{2}$ and $-\sqrt{2}$ are indistinguishable from $\mathbb{Q}$. So they can be swapped without $\mathbb{Q}$ noticing. All such swaps form the Galois group.

$$x^2 - 2 = 0$$

roots: $\sqrt{2}$, $-\sqrt{2}$

$$\text{Gal}(\mathbb{K}/\mathbb{Q}) = \{ e, (1\ 2) \}$$

order-2 group swapping $\sqrt{2}$ and $-\sqrt{2}$.
 $\mathbb{K}=\mathbb{Q}(\sqrt{2})$.

$$x^3 - 1 = 0$$

roots: 1 , ω , ω^2

$$\text{Gal}(\mathbb{K}/\mathbb{Q}) = \{ e, (\omega\ \omega^2) \}$$

1 is known; the unknowns ω , ω^2 give order 2.

$$x^3 - 2 = 0$$

roots: α , $\alpha\omega$, $\alpha\omega^2$ ($\alpha=\sqrt[3]{2}$)

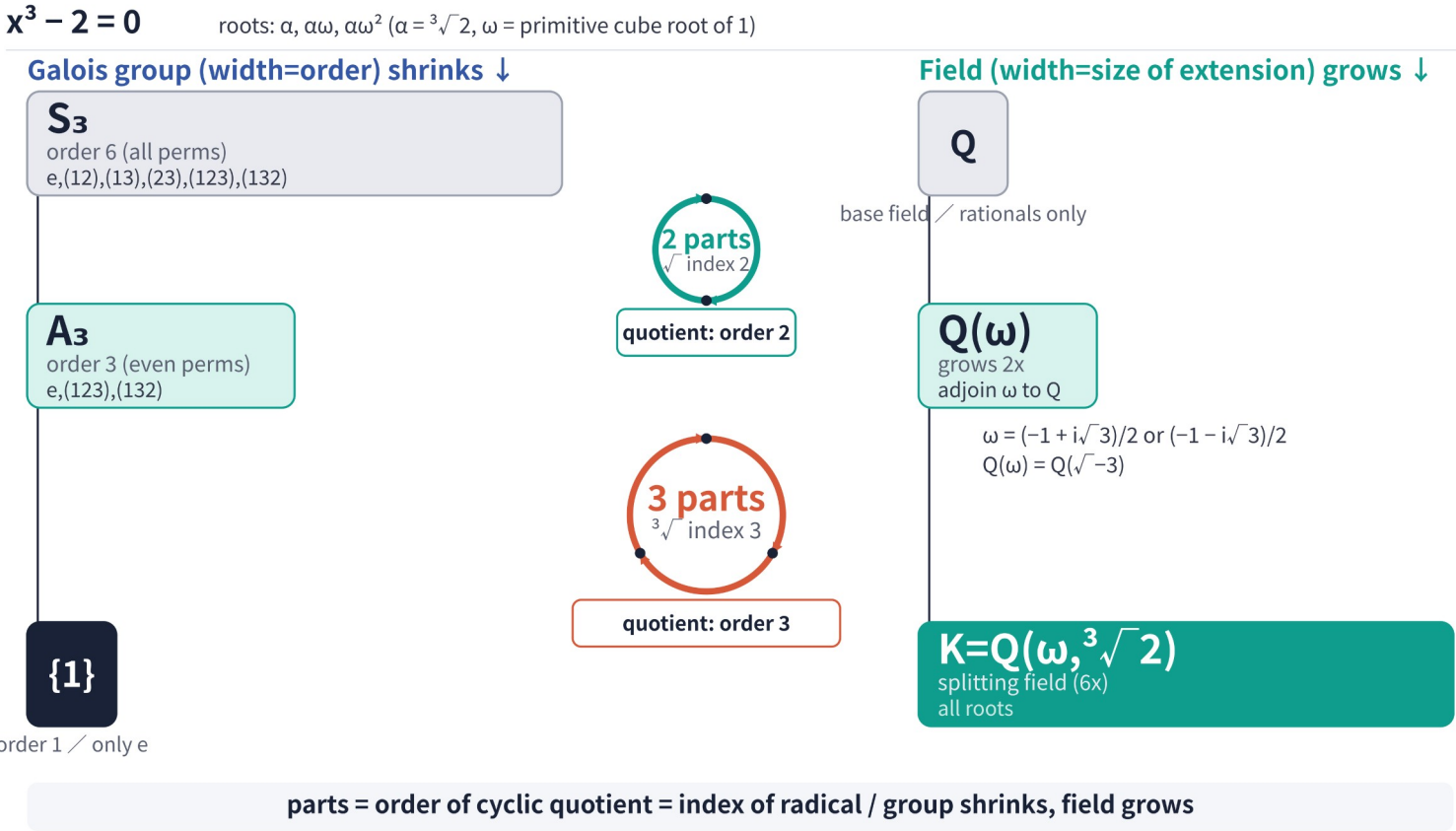
$$\text{Gal}(\mathbb{K}/\mathbb{Q}) = S_3$$

adjoin two things: ω (contains imaginary $\sqrt{-3}$) and a cube root.

● Appendix 2

Image: one turn — the group shrinks and the field grows

The order of the quotient equals the number of permutation steps needed to cycle once through the radical's conjugate candidates and return — i.e. the figure's number of “parts.”



● Appendix 2

The Adjoined Element's Conjugates and the Cyclic Quotient's Generator Correspond

* Assumes a single-step radical extension (adjoining one radical extends the field by one step)

Assume: $L = M(\alpha)$, $\alpha^n \in M$, $\zeta_n \in M$
Then $\text{Gal}(L/M)$ is the cyclic group C_n , with generator σ satisfying $\sigma(\alpha) = \zeta_n \alpha$

Rotation of the Conjugates

The generator σ advances one step at a time
 $\sigma(\zeta^k \alpha) = \zeta^{k+1} \alpha$
(indices computed mod n)

1-to-1

Each coset $g^k H$ corresponds to the action sending α to $\zeta^k \alpha$

α	$\leftrightarrow$	H
$\zeta \alpha$	$\leftrightarrow$	gH
$\zeta^2 \alpha$	$\leftrightarrow$	$g^2 H$
$\vdots$	$\leftrightarrow$	$\vdots$
$\zeta^{n-1} \alpha$	$\leftrightarrow$	$g^{n-1} H$

Cycling of the Quotient Group

$G/H = \langle gH \rangle \cong C_n$
($G = \text{Gal}(L/M)$, $H = \{\text{id}\}$)

What Is the 1-to-1 Correspondence, Exactly?

The coset $g^k H$ corresponds to α 's image $\zeta^k \alpha$

$(g^k H)(\alpha) = \zeta^k \alpha$

Summary:

- ① The generator σ : $\alpha \mapsto \zeta \alpha$ cycles the conjugates.
- ② The generator gH cycles the quotient group G/H .
- ③ Its k -th power sends α to $\zeta^k \alpha$ — the same cyclic structure appears both ways.

● Appendix 2

The shrinking chain for x^3-2 and the solvability conclusion

Two steps to solve $x^3-2=0$

STEP 1: adjoin ω

field: $Q \rightarrow L = Q(\omega)$

group: $S_3 \rightarrow A_3$ (order 3)

quotient order 2 = square root ($\sqrt{-3}$)

STEP 2: adjoin $\sqrt[3]{2}$

field: $L \rightarrow K = Q(\omega, \sqrt[3]{2})$

group: $A_3 \rightarrow \{e\}$

quotient order 3 = cube root

Once the group shrinks to $\{e\}$, no unknowns remain and the adjoined elements express the roots.

Generalization: what "solvable by radicals" means

At each step, the quotient of the larger group by the smaller (normal) subgroup must be cyclic. The order of that cyclic group equals the index of the radical adjoined at that step.

Being solvable by radicals means this shrinking-and-growing repeats until the group reaches $\{e\}$ and the field K reaches the splitting field.

Up to degree 4, such a chain exists for any coefficients, so all are solvable by radicals. From degree 5 up, some equations have no such chain; those that cannot reach $\{e\}$ are not solvable by radicals.

Common early stumbling points (1/2)

1 Roots are not missing — they may just be inexpressible by radicals [\(main → P9\)](#)

For degree ≥ 5 , roots always exist if we allow complex numbers. The question is not whether roots exist, but whether they can be written using only **the four operations and radicals**.

"No solution formula" does not mean there are no roots — it means no uniform procedure expresses the roots by radicals alone.

2 Some individual quintics are solvable [\(main → P9\)](#)

"No formula for the quintic" does not mean every quintic is unsolvable. Some quintics are solvable by radicals (e.g. $x^5 - 1 = 0$).

The point is that for the **general quintic with literal coefficients**, no uniform formula in the four operations and radicals exists.

3 The Galois group is not all permutations of the roots [\(main → P3\)](#)

The Galois group is not every rearrangement of the roots, but only **the swaps that preserve the value of rational expressions built from the roots**.

Roots indistinguishable from the rationals may be swapped, but swaps that break relations like sums or products are not allowed.

● Appendix 3

Common early stumbling points (2/2)

4

"The group shrinks" means fewer allowed swaps (main → P4)

Adjoining a new element to the field makes previously indistinguishable numbers distinguishable, so fewer swaps are allowed.
field grows → more known numbers → more numbers to fix → fewer allowed swaps → the Galois group shrinks.

5

The quotient's order corresponds to which n-th root is adjoined (main → P4 • App.2 P15)

The quotient's order equals the number of permutation steps to cycle once through the radical's conjugate candidates and return.

For a square root the candidates are $\sqrt{a}$, $-\sqrt{a}$ (2 of them): $\sqrt{a} \rightarrow -\sqrt{a} \rightarrow \sqrt{a}$ returns after 2 steps.

For a cube root the candidates are $\sqrt[3]{a}$, $\omega\sqrt[3]{a}$, $\omega^2\sqrt[3]{a}$ (3 of them): $\sqrt[3]{a} \rightarrow \omega\sqrt[3]{a} \rightarrow \omega^2\sqrt[3]{a} \rightarrow \sqrt[3]{a}$ returns after 3 steps.

6

The adjoined element is not always a root itself (App.1 P13 • App.2 P15)

The element adjoined when shrinking is not necessarily a root of the equation; a helper number needed to express the roots may be adjoined first.

Example: in $ax^2+bx+c=0$, the $\sqrt{b^2-4ac}$ in the quadratic formula is not a root itself but a radical adjoined to express the roots.

For x^3-2 too, among the roots $\alpha, \alpha\omega, \alpha\omega^2$ ($\alpha=\sqrt[3]{2}$), ω is not a root, yet expressing all three needs ω .

Solving by radicals = adjoining the needed numbers in order until the group shrinks to $\{e\}$.